

# ARDL Summary from RW4 Primary Chain

## INTERPRETATION GUIDE

- ADF  $p > 0.05$  and KPSS  $p < 0.05$  -> likely  $I(1)$ -like; prefer differences/cointegration models.
- ADF  $p < 0.05$  and KPSS  $p > 0.05$  -> stationary; level models are more admissible.
- Ljung-Box  $p < 0.05$  -> autocorrelation; add lag structure.
- BP/White  $p < 0.05$  -> heteroskedasticity; use robust or HAC standard errors.
- JB  $p < 0.05$  -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

## RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end.

family=ARDL

rows=279

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk\_Dairy\_Research/outputs/primary\_chain\_summary/primary\_chain\_consolidated.xlsx

Table: ARDL\_Summary

| panel_level | panel_name                   | segment_name            | product                 | standardized_type    | retailer_panel          | brand | price_variant | reconstruction_variant | farm_gate | frequency | chain_direction | stage_from | stage_to   | link                     | model_family | y_series   | x_series   | n_obs | sr_coef                | lr_coef               | ect_coef | ect_pvalue | asymmetry_short_p | asymmetry_long_p | vecm_rank  | model_status | unreliable_flag           | unreliable_reason | shock_dummy_included | promo_controls_included | ljungbox_p                   | arch_p                       | jb_p                         | notes | robust_linear_vs_pchip | interpolation_sensitive | robust_across_reconstruction | core_finding_flag |
|-------------|------------------------------|-------------------------|-------------------------|----------------------|-------------------------|-------|---------------|------------------------|-----------|-----------|-----------------|------------|------------|--------------------------|--------------|------------|------------|-------|------------------------|-----------------------|----------|------------|-------------------|------------------|------------|--------------|---------------------------|-------------------|----------------------|-------------------------|------------------------------|------------------------------|------------------------------|-------|------------------------|-------------------------|------------------------------|-------------------|
| product     | product::Вершки::Silpo::o... | Вершки                  | Вершки                  | cream                | Silpo                   | nan   | observed      | linear                 | initial   | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | ARDL         | Retail     | ProZorro   | 41    | 0.03815343029021434    | -0.07780863784892769  | nan      | nan        | nan               | nan              | nan        | ok           | 0                         | nan               | 0                    | 0.8780631987617009      | 0.2706901208918511           | 0.65770037746761             | bounds_lower_p=0.0001846;... | 1     | 0                      | 0                       | 0                            |                   |
| product     | product::Вершки::Silpo::o... | Вершки                  | Вершки                  | cream                | Silpo                   | nan   | observed      | linear                 | initial   | daily     | forward         | FarmGateUA | ProZorro   | farmgateua_to_prozorro   | ARDL         | ProZorro   | FarmGateUA | 41    | 27.40581164655441      | -1.563367219701413    | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.04265890648236325  | 0.947046008010218       | 0.1379433157208637           | bounds_lower_p=0.5287; bo... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Молоко питне::Si... | Молоко питне            | Молоко питне            | milk                 | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | Retail     | ProZorro   | retail_to_prozorro       | ARDL         | ProZorro   | Retail     | 41    | -1.569298877962051     | 1.462494024258335     | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8435337514070991   | 0.9672870409827264      | 1.494891740049852e-10        | bounds_lower_p=0.0002924;... | 0                            | 0     | 0                      | 0                       |                              |                   |
| product     | product::Молоко питне::Si... | Молоко питне            | Молоко питне            | milk                 | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | ProZorro   | ProducerUA | prozorro_to_producerua   | ARDL         | ProducerUA | ProZorro   | 41    | -0.003981249234879858  | 0.08556001807837454   | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.0196200671200218   | 0.2375437563121033      | 0.7892194337359504           | bounds_lower_p=0.004736;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Молоко питне::Si... | Молоко питне            | Молоко питне            | milk                 | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | Retail     | ProducerUA | retail_to_producerua     | ARDL         | ProducerUA | Retail     | 41    | -0.01904566173387146   | 0.1126183560663258    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.3474130360123434   | 0.925869723100835       | 3.698771932262856e-05        | bounds_lower_p=2.97e-07;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | observed      | linear                 | initial   | daily     | forward         | ProducerUA | ProZorro   | producerua_to_prozorro   | ARDL         | ProZorro   | ProducerUA | 38    | 2.533791668492441      | 0.8515901678330565    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8950671950008878   | 0.7500372663994986      | 0.02058974429894947          | bounds_lower_p=6.579e-08;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | observed      | linear                 | initial   | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | ARDL         | Retail     | ProZorro   | 38    | -0.008082951427838936  | 0.1583981708763298    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.7406617844250787   | 0.6046290618248675      | 0.8118949225796651           | bounds_lower_p=0.9132; bo... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | observed      | linear                 | initial   | daily     | forward         | FarmGateUA | Retail     | farmgateua_to_retail     | ARDL         | Retail     | FarmGateUA | 38    | -16.12941410918768     | -3.273094954784757    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8090316794651169   | 0.5742982621103152      | 0.05595308976443426          | bounds_lower_p=0.7505; bo... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | ProZorro   | ProducerUA | prozorro_to_producerua   | ARDL         | ProducerUA | ProZorro   | 38    | 0.05875093381259333    | -0.0758952357905691   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8636489987653581   | 0.4300526930049906      | 0.1580768609460545           | bounds_lower_p=0.03344; b... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | ARDL         | FarmGateUA | Retail     | 38    | -0.004884939285385581  | 0.008156888394341669  | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.08807849703846922  | 0.3329413806206964      | 0.1521775431549344           | bounds_lower_p=6.961e-08;... | 0                            | 0     | 0                      | 0                       |                              |                   |
| product     | product::Сметана::Silpo::... | Сметана                 | Сметана                 | sour_cream           | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | ProZorro   | FarmGateUA | prozorro_to_farmgateua   | ARDL         | FarmGateUA | ProZorro   | 38    | -0.0002561523368839191 | -0.007181451310785285 | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.7838005674024638   | 0.002032162462147065    | 7.447112047328443e-08        | bounds_lower_p=0.004736;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сметана::Silpo::... | Сметана                 | Сметана                 | sour_cream           | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | ARDL         | FarmGateUA | Retail     | 38    | 0.001083444100194342   | -0.1056337246281378   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.7373514597990629   | 0.6232340801443153      | 0.9393878293158436           | bounds_lower_p=0.001988;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Вершки::Silpo::b... | Вершки                  | Вершки                  | cream                | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | FarmGateUA | ProZorro   | farmgateua_to_prozorro   | ARDL         | ProZorro   | FarmGateUA | 41    | 27.40581164655441      | -1.563367219701413    | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.04265890648236325  | 0.947046008010218       | 0.1379433157208637           | bounds_lower_p=0.5287; bo... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Вершки::Silpo::b... | Вершки                  | Вершки                  | cream                | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | ARDL         | FarmGateUA | Retail     | 41    | -0.00635974347776247   | 0.1331639064336117    | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.9330608628619284   | 0.0255405043019595      | 0.9504050602483913           | bounds_lower_p=0.001505;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Молоко питне::Si... | Молоко питне            | Молоко питне            | milk                 | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | ARDL         | Retail     | ProZorro   | 41    | -0.03270184240215679   | -0.1549879499573674   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.238498861809109    | 0.6226307898142169      | 0.1274605250964556           | bounds_lower_p=9.474e-08;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Молоко питне::Si... | Молоко питне            | Молоко питне            | milk                 | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | ProZorro   | ProducerUA | prozorro_to_producerua   | ARDL         | ProducerUA | ProZorro   | 41    | -0.003981249234879858  | 0.08556001807837454   | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.0196200671200218   | 0.2375437563121033      | 0.7892194337359504           | bounds_lower_p=0.004736;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | ProducerUA | ProZorro   | producerua_to_prozorro   | ARDL         | ProZorro   | ProducerUA | 38    | 2.533791668492441      | 0.8515901678330565    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8950671950008878   | 0.7500372663994986      | 0.02058974429894947          | bounds_lower_p=6.579e-08;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | ARDL         | Retail     | ProZorro   | 38    | -0.2316252020709719    | 0.08260811044262882   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.3647034991231822   | 0.9328813954406262      | 0.162951764020707            | bounds_lower_p=0.0002747;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | ProducerUA | Retail     | producerua_to_retail     | ARDL         | Retail     | ProducerUA | 38    | 1.860159048436145      | -7.658750575624781    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.73178223354916687  | 0.6411449709153603      | 0.6848535936402766           | bounds_lower_p=2.57e-08;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | FarmGateUA | Retail     | farmgateua_to_retail     | ARDL         | Retail     | FarmGateUA | 38    | 14.40021465960636      | -1.000896839545215    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.44982137565602643  | 0.6367138733015969      | 0.02702673654317345          | bounds_lower_p=4.74e-05;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | ProZorro   | ProducerUA | prozorro_to_producerua   | ARDL         | ProducerUA | ProZorro   | 38    | 0.05875093381259333    | -0.0758952357905691   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8636489987653581   | 0.4300526930049906      | 0.1580768609460545           | bounds_lower_p=0.03344; b... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | Retail     | ProducerUA | retail_to_producerua     | ARDL         | ProducerUA | Retail     | 38    | 0.1199315068832618     | -0.06428989223278714  | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8318090481054523   | 0.8945663500866514      | 0.03200554792406992          | bounds_lower_p=0.01982; b... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сир твердий::Sil... | Сир твердий             | Сир твердий             | hard_cheese          | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | ARDL         | FarmGateUA | Retail     | 38    | 0.01173175474368326    | 0.5752798070776506    | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.080765103651483    | 0.03046060817270044     | 0.000592702044422381         | bounds_lower_p=3.795e-10;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| product     | product::Сметана::Silpo::... | Сметана                 | Сметана                 | sour_cream           | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | ProZorro   | FarmGateUA | prozorro_to_farmgateua   | ARDL         | FarmGateUA | ProZorro   | 38    | -0.0002561523368839191 | -0.007181451310785285 | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.7838005674024638   | 0.002032162462147065    | 7.447112047328443e-08        | bounds_lower_p=0.004736;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::observed:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | observed      | linear                 | initial   | daily     | forward         | ProZorro   | Retail     | prozorro_to_retail       | ARDL         | Retail     | ProZorro   | 47    | 0.001079202784507904   | 0.01230010706723881   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.9504865381966955   | 0.7111529281299548      | 0.62124699989172789          | bounds_lower_p=0.05069; b... | 0                            | 0     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::observed:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | ProducerUA | FarmGateUA | producerua_to_farmgateua | ARDL         | FarmGateUA | ProducerUA | 47    | -0.02058815029147709   | 0.02026681966645144   | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.346451535684855    | 0.00286418829106341     | 2.705836385479854e-15        | bounds_lower_p=6.048e-06;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::observed:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | ProZorro   | FarmGateUA | prozorro_to_farmgateua   | ARDL         | FarmGateUA | ProZorro   | 47    | -0.0002567751815376607 | 0.00825046634993702   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8781686329517566   | 0.01062452267357634     | 0.01062452267357634          | bounds_lower_p=8.465e-08;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::observed:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | observed      | linear                 | initial   | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | ARDL         | FarmGateUA | Retail     | 47    | 0.0008968178666250992  | -0.08909618273334395  | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8404752054940611   | 0.8674521940346733      | 0.6542306273508575           | bounds_lower_p=1.11e-16;...  | 0                            | 0     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::baseline:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | forward         | ProducerUA | Retail     | producerua_to_retail     | ARDL         | Retail     | ProducerUA | 47    | 0.6431587542903205     | -0.1566372973389772   | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.6562879182615633   | 0.950280320107265       | 0.7793703614095745           | bounds_lower_p=0.000605;...  | 0                            | 1     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::baseline:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | Retail     | ProducerUA | retail_to_producerua     | ARDL         | ProducerUA | Retail     | 47    | 0.1177788039343319     | -3.42026337340533     | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.70797282540931078  | 3.58665648200218e-07    | bounds_lower_p=0.8095; bo... | 0                            | 1                            | 0     | 0                      |                         |                              |                   |
| average     | average::Silpo::baseline:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | ProducerUA | FarmGateUA | producerua_to_farmgateua | ARDL         | FarmGateUA | ProducerUA | 47    | -0.02058815029147709   | 0.02026681966645144   | nan      | nan        | nan               | nan              | unreliable | 1            | residual_autocorr_or_arch | 0                 | 0.346451535684855    | 0.00286418829106341     | 2.705836385479854e-15        | bounds_lower_p=6.048e-06;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::baseline:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | ProZorro   | FarmGateUA | prozorro_to_farmgateua   | ARDL         | FarmGateUA | ProZorro   | 47    | -0.0002567751815376607 | -0.00825046634993702  | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.8781686329517566   | 0.01062452267357634     | 0.01062452267357634          | bounds_lower_p=8.465e-08;... | 0                            | 1     | 0                      | 0                       |                              |                   |
| average     | average::Silpo::baseline:... | all_products_average    | all_products_average    | all_products_average | Silpo                   | nan   | baseline      | linear                 | initial   | daily     | reverse         | Retail     | FarmGateUA | retail_to_farmgateua     | ARDL         | FarmGateUA | Retail     | 47    | 0.000263196846095326   | 0.3674418421843634    | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.5310242615301275   | 0.1690261871606671      | 5.104976966727216e-05        | bounds_lower_p=1.597e-06;... | 0                            | 0     | 0                      | 0                       |                              |                   |
| comparison  | comparison::silpo_vs_novu... | silpo_vs_novus_observed | silpo_vs_novus_observed | comparison_panel     | silpo_vs_novus_observed | nan   | observed      | linear                 | initial   | daily     | forward         | ProducerUA | ProZorro   | producerua_to_prozorro   | ARDL         | ProZorro   | ProducerUA | 49    | 1.401469844673399      | 1.261386920487273     | nan      | nan        | nan               | nan              | ok         | 0            | nan                       | 0                 | 0.348439168993778    | 0.1814822357263797      | 0.6834057185870              |                              |                              |       |                        |                         |                              |                   |